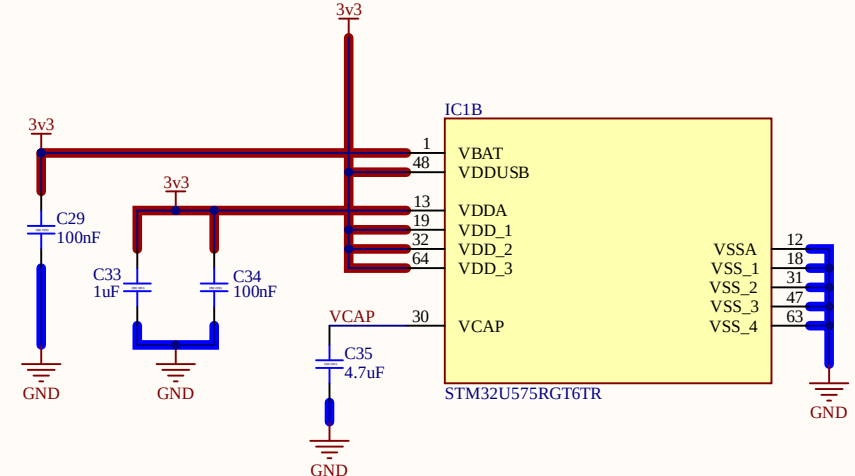
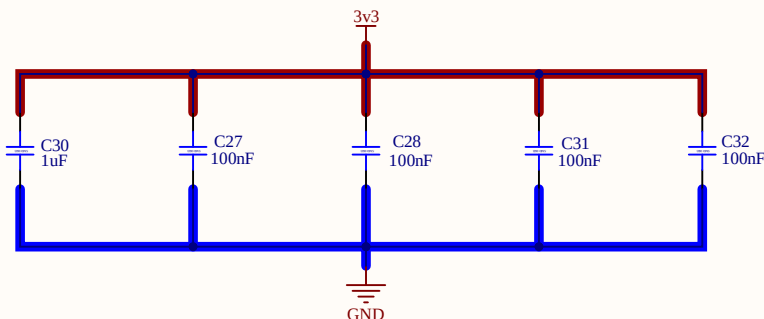
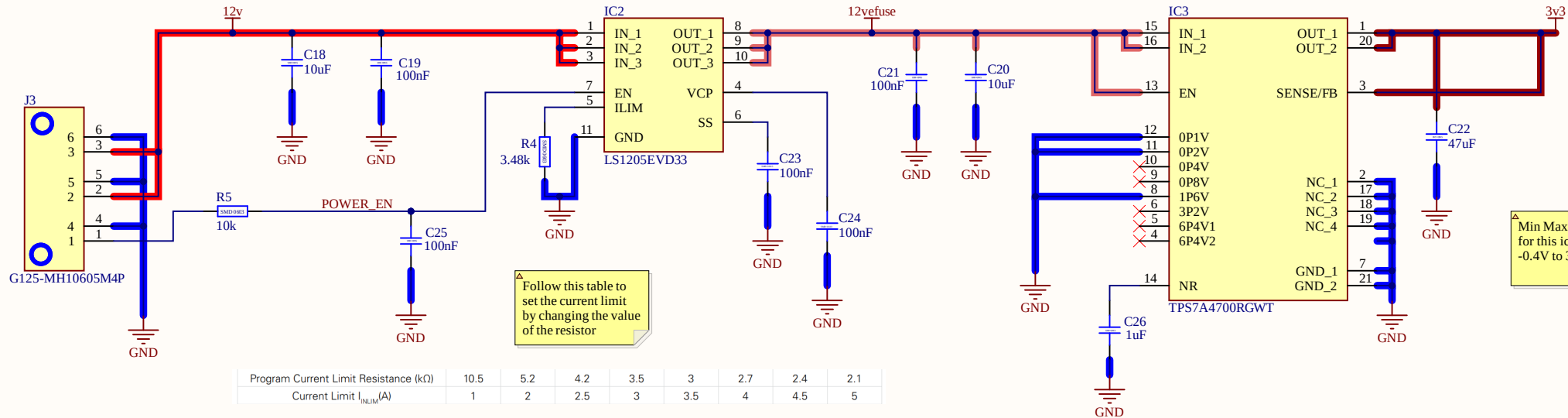
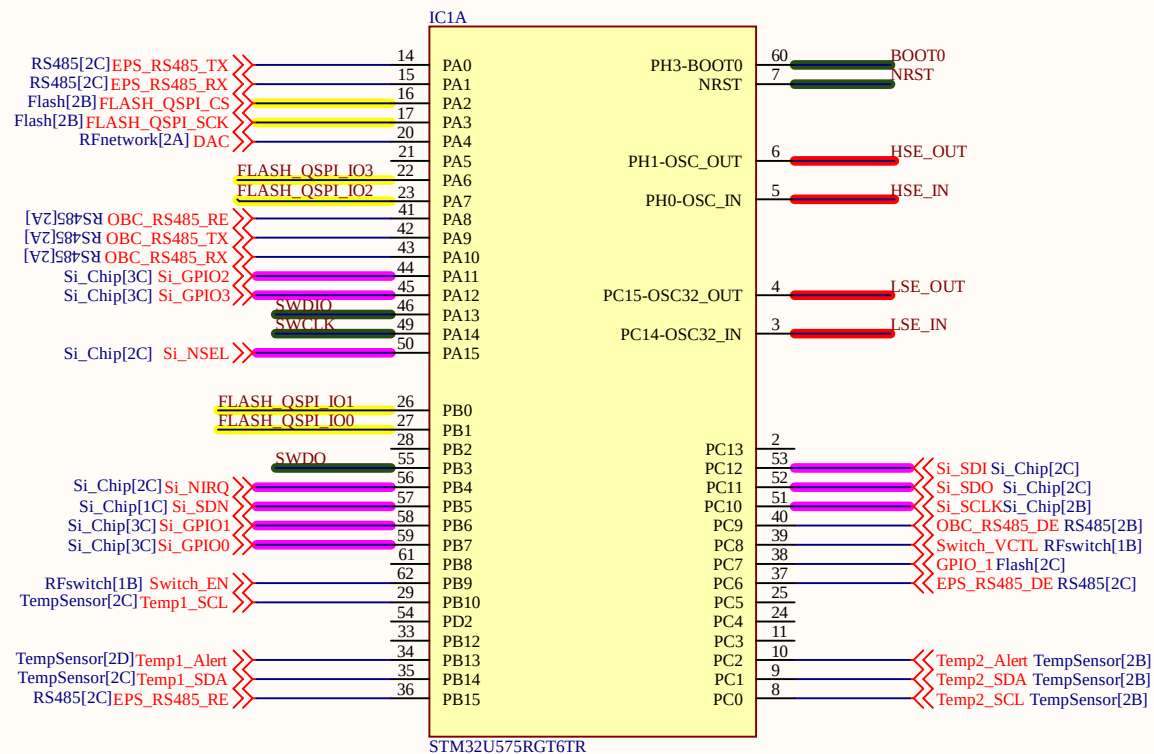
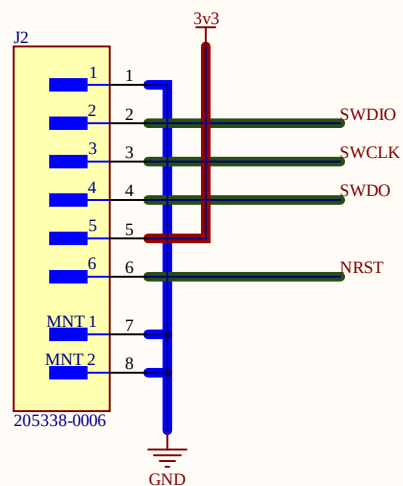
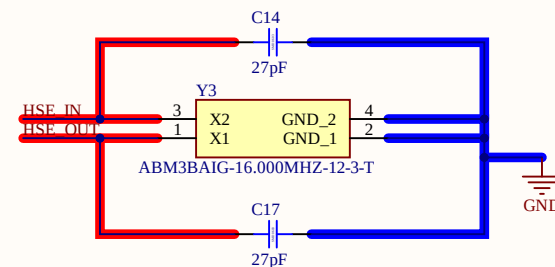
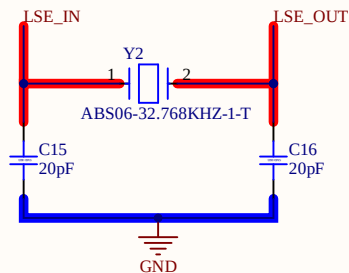
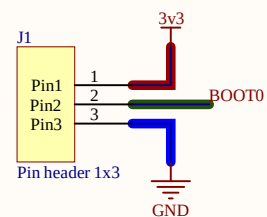




POWER

Title		
Size	Number	Revision
A4		
Date:	3-26-2026	Sheet of
File:	C:\Users\...\Power.SchDoc	Drawn By:



STM32

Title

Size

A4

Number

Revision

Date:

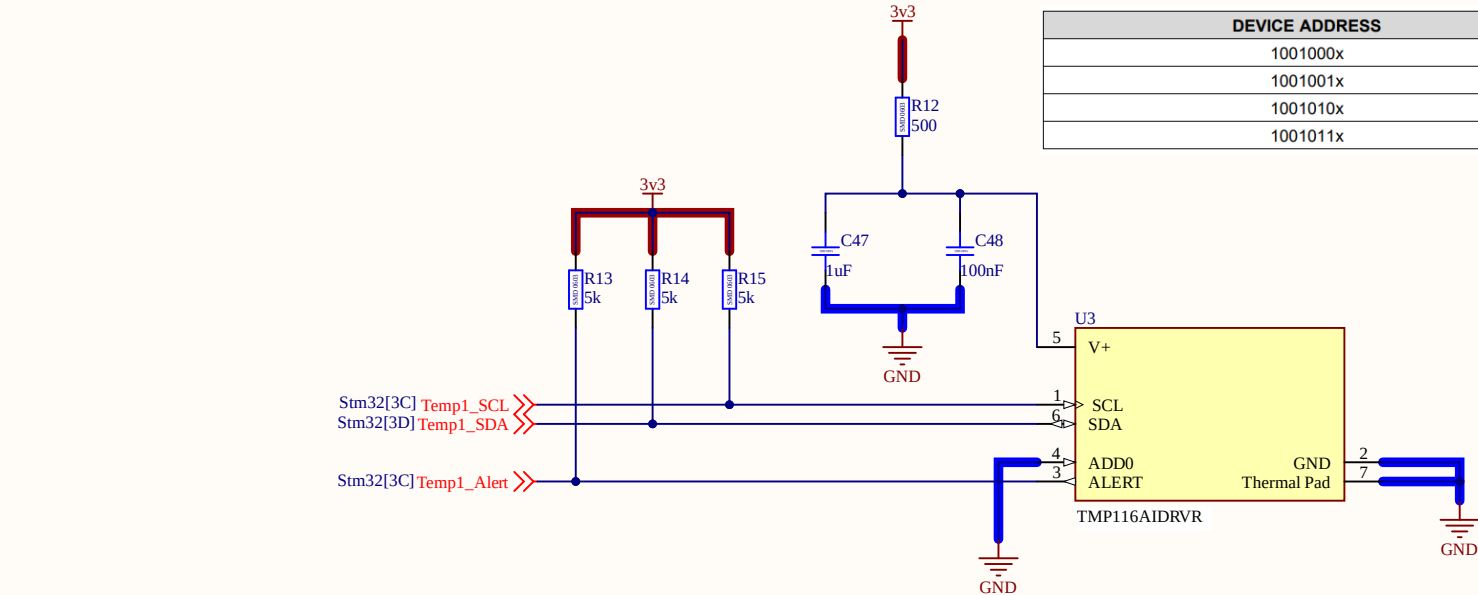
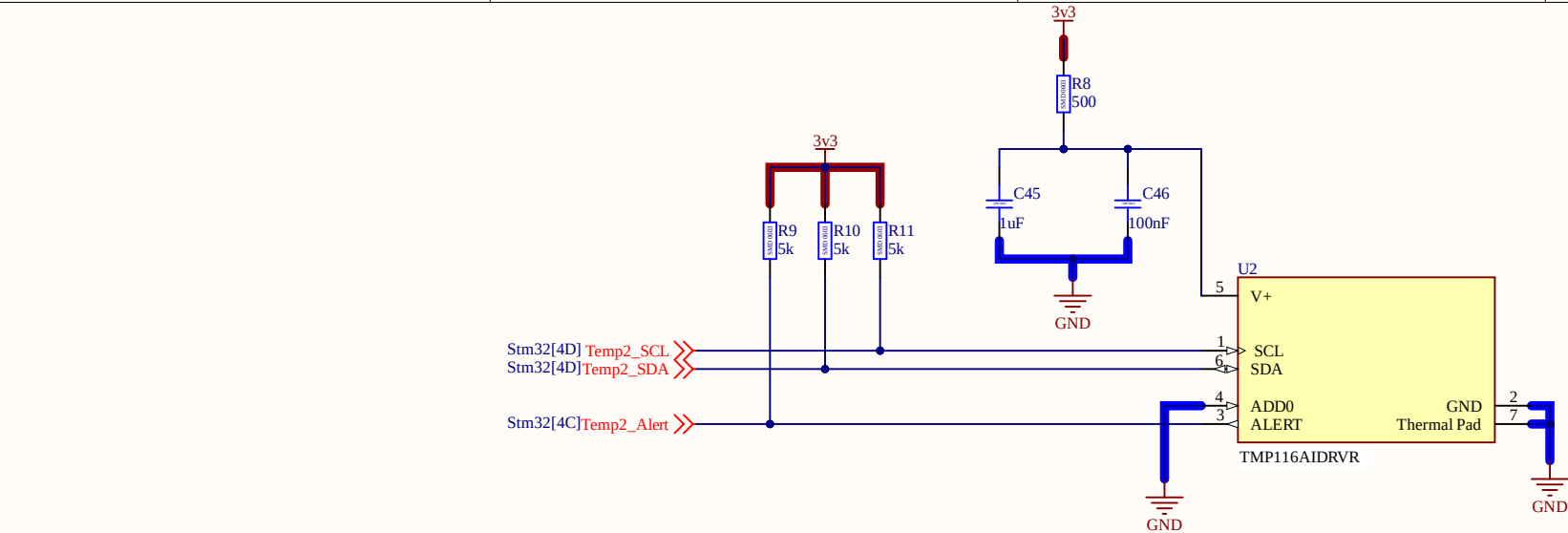
3-26-2026

Sheet of

File:

C:\Users\...\Stm32.SchDoc

Drawn By:



DEVICE ADDRESS	ADD0 PIN CONNECTION
1001000x	Ground
1001001x	V+
1001010x	SDA
1001011x	SCL

TEMP SENSOR

Title		
Size A4	Number	Revision
Date: 3-26-2026	Sheet of	
File: C:\Users\...\TempSensor.SchDoc	Drawn By:	

A

A

B

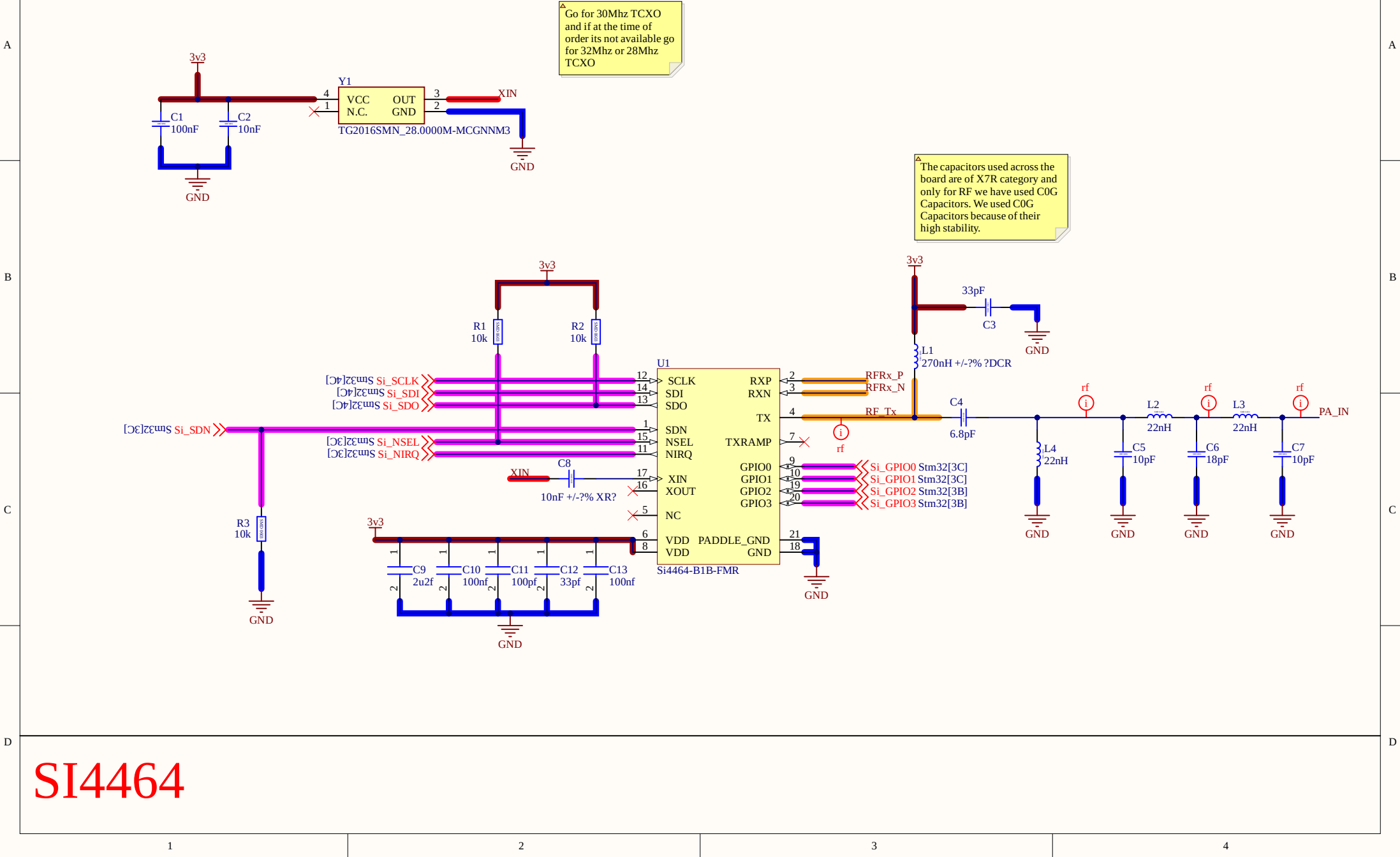
B

C

C

D

D



1

2

3

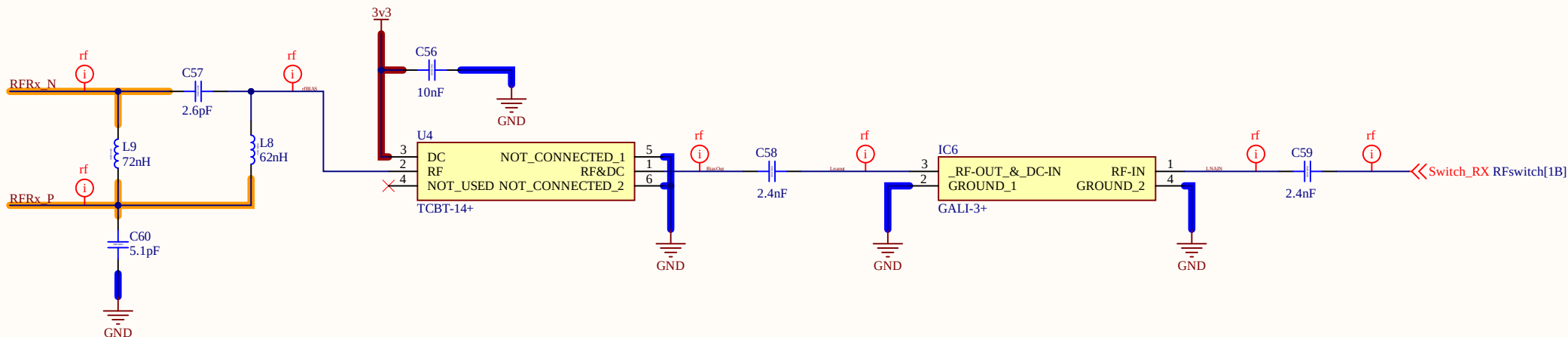
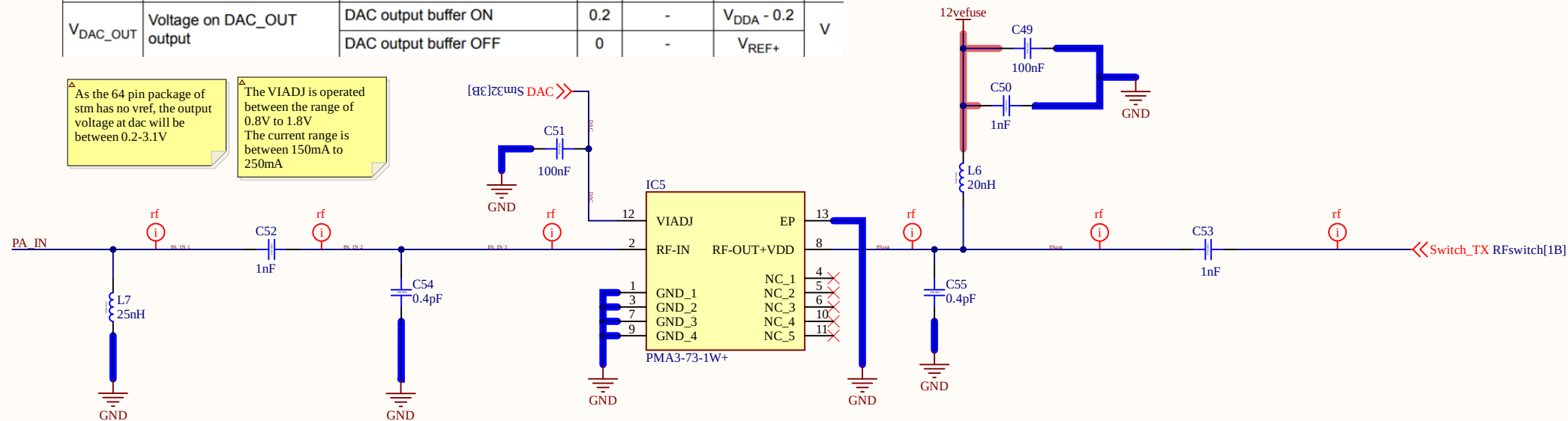
4

V _{DAC_OUT}	Voltage on DAC_OUT output	DAC output buffer ON	0.2	-	V _{DDA} - 0.2	V
		DAC output buffer OFF	0	-	V _{REF+}	

As the 64 pin package of stm has no vref, the output voltage at dac will be between 0.2-3.1V

The VIADJ is operated between the range of 0.8V to 1.8V
The current range is between 150mA to 250mA

[[[323B]]] DAC



RF CIRCUIT

Title

Size

A4

Number

Revision

Date: 3-26-2026

Sheet of

File: C:\Users\RFNetwork.SchDoc

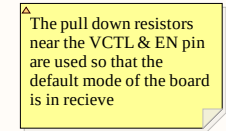
Drawn By:

1

2

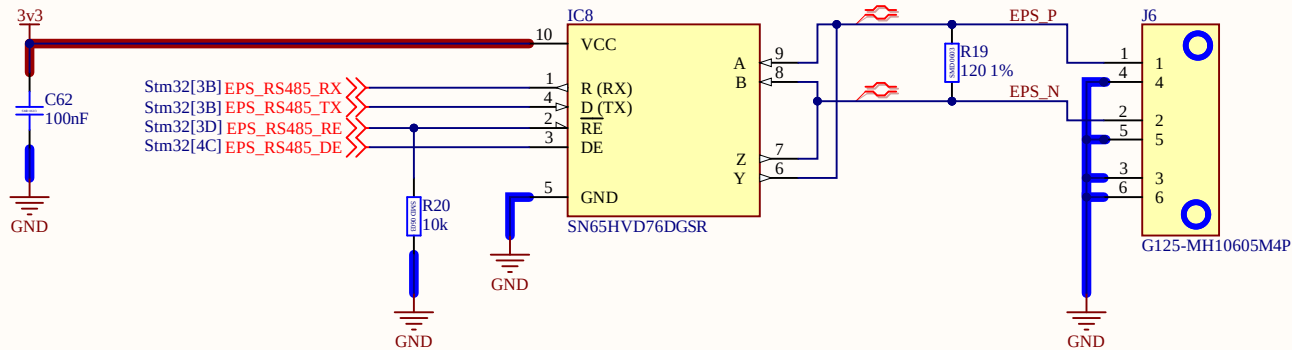
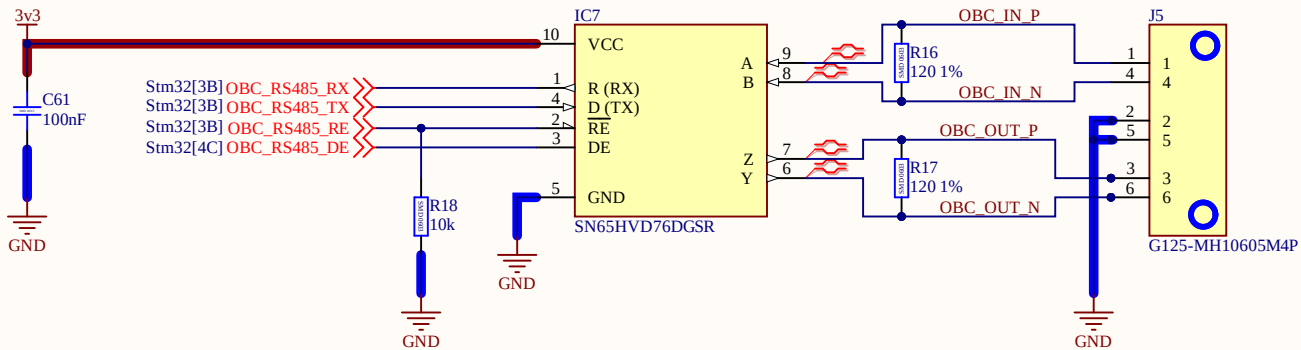
3

4



Control Input		Signal Path State	
V _{CTL} State	EN State	RFC to RF1	RFC to RF2
Low	Low	Off	On
High	Low	On	Off
Low	High	Off	Off
High	High	Off	Off

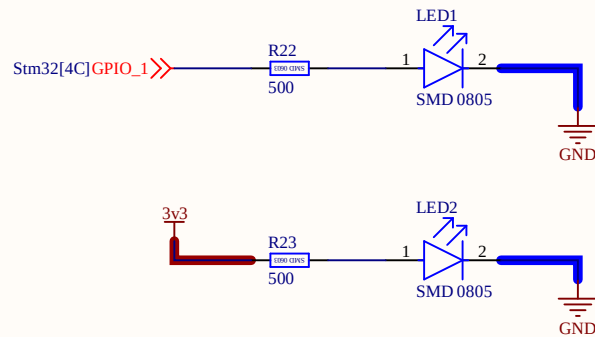
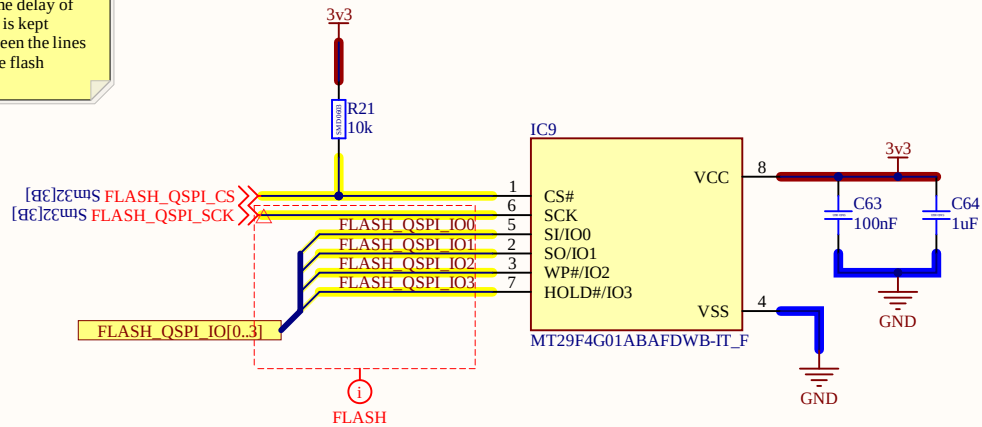
Title		
Size A4	Number	Revision
Date:	3-26-2026	Sheet of
File:	C:\Users\...\RFswitch.SchDoc	Drawn By:



RS485

△ A NAND Flash Memory is used because of its high density to store data

△ A time delay of 20ps is kept between the lines of the flash



FLASH MEMORY

Title		
Size A4	Number	Revision
Date: 3-26-2026	Sheet of	
File: C:\Users\Flash.SchDoc	Drawn By:	



Power Indicator

PUT IT BACK

FLASH

OBC RS485

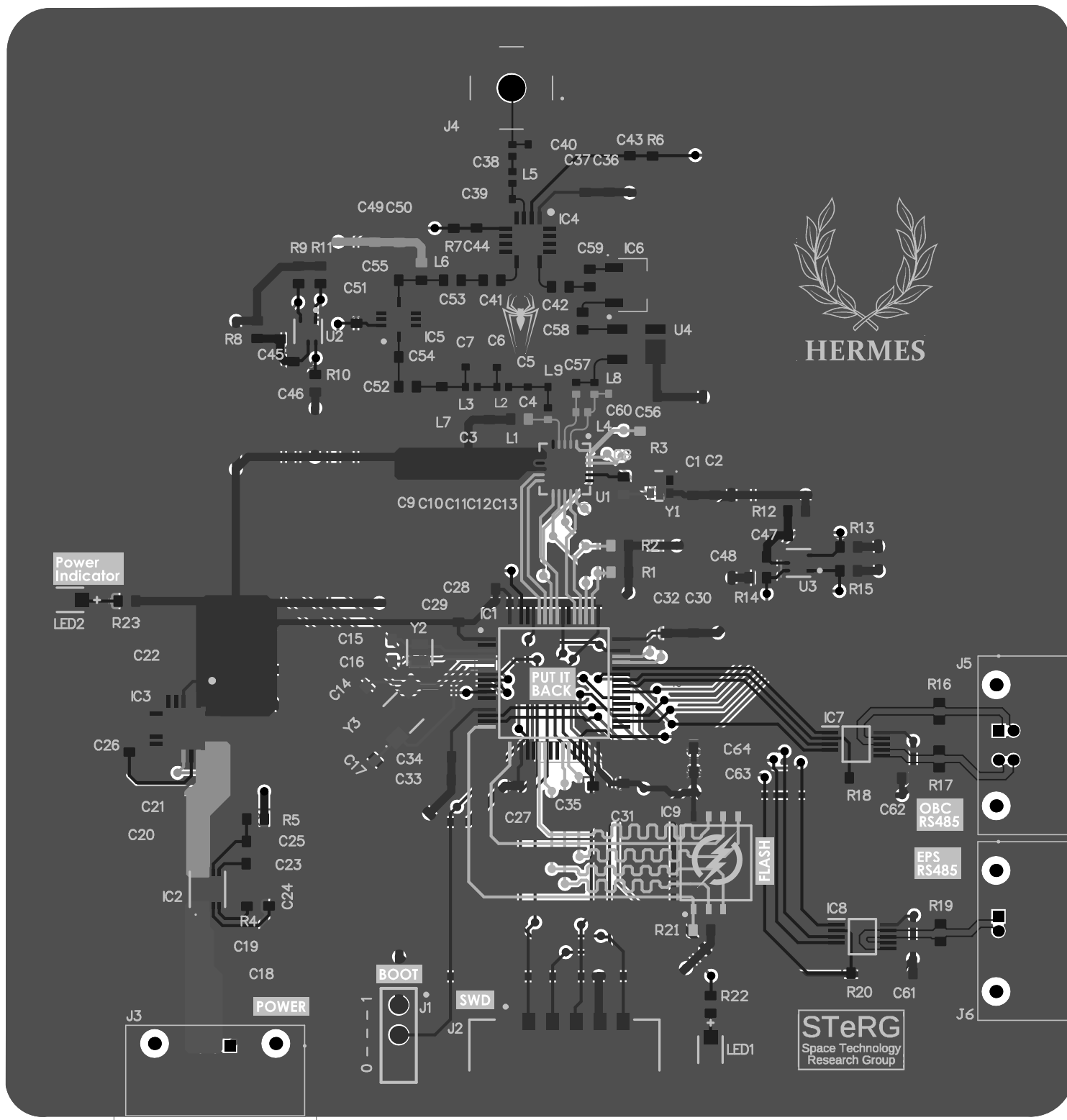
EPS RS485

BOOT

SWD

POWER

STeRG
Space Technology
Research Group



Comment	Description	Designator	Footprint	LibRef	Quantity
100nF	Capacitor SMD 0603 RATED	C1, C19, C21, C23, C24, C25, C27, C28, C29, C31, C32, C34, C36, C46, C48, C49, C51, C61, C62, C63	CAP_0603	Capacitor-SMD-0603	20
10nF	Capacitor SMD 0603 RATED	C2, C56	CAP_0603	Capacitor-SMD-0603	2
33pF	Capacitor SMD 0603 RATED	C3	CAP_0603	Capacitor-SMD-0603	1
6.8pF	Capacitor SMD 0402 RATED	C4	CAP_0402	Capacitor-SMD-0402	1
10pF	Capacitor SMD 0402 RATED	C5, C7	CAP_0402	Capacitor-SMD-0402	2
18pF	Capacitor SMD 0402 RATED	C6	CAP_0402	Capacitor-SMD-0402	1
10nF +/-1% XR7	Capacitor SMD 0603 RATED	C8	CAP_0603	Capacitor-SMD-0603	1
2u2f	Capacitor	C9	CAPC1608X87N	0603N470500CT	1
100nF	Capacitor	C10, C13	CAPC1608X87N	0603N470500CT	2
100pF	Capacitor	C11	CAPC1608X87N	0603N470500CT	1
33pF	Capacitor	C12	CAPC1608X87N	0603N470500CT	1
27pF	Capacitor SMD 0603 RATED	C14, C17	CAP_0603	Capacitor-SMD-0603	2
20pF	Capacitor SMD 0603 RATED	C15, C16	CAP_0603	Capacitor-SMD-0603	2
10uF	Capacitor SMD 0805 RATED	C18, C20	CAP_0805	Capacitor-SMD-0805	2
47uF	Capacitor SMD 1210 RATED	C22	CAP_1210	Capacitor-SMD-1210	1
1uF	Capacitor SMD 0603 RATED	C26, C30, C33, C45, C47, C64	CAP_0603	Capacitor-SMD-0603	6
4.7uF	Capacitor SMD 0603 RATED	C35	CAP_0603	Capacitor-SMD-0603	1
100pF	Capacitor SMD 0603 RATED	C37, C41, C42, C43, C44	CAP_0603	Capacitor-SMD-0603	5
6.4pF	Capacitor SMD 0402 RATED	C38	CAP_0402	Capacitor-SMD-0402	1
2.6pF	Capacitor SMD 0402 RATED	C39, C40, C57	CAP_0402	Capacitor-SMD-0402	3
1nF	Capacitor SMD 0603 RATED	C50, C52, C53	CAP_0603	Capacitor-SMD-0603	3
0.4pF	Capacitor SMD 0603 RATED	C54, C55	CAP_0603	Capacitor-SMD-0603	2
2.4nF	Capacitor SMD 0603 RATED	C58, C59	CAP_0603	Capacitor-SMD-0603	2
5.1pF	Capacitor SMD 0402 RATED	C60	CAP_0402	Capacitor-SMD-0402	1
STM32U575RG16TR	Integrated Circuit	IC1	QFP50P1200X1200X1 60-64N	STM32U575RG16TR	1
LS1205EVD33	Integrated Circuit	IC2	LS1205EVD33	LS1205EVD33	1
TPS7A4700RGWT	Integrated Circuit	IC3	QFN65P500X500X100- 21N-D	TPS7A4700RGWR	1
HMC8038LP4CETR	Integrated Circuit	IC4	QFN65P400X400X90- 17N-D	HMC8038LP4CETR	1
PM43-73-1W+	Integrated Circuit	IC5	QFN51P300X300X89- 13N	PM43-73-1W+	1
GALI-3+	Integrated Circuit	IC6	GALI3	GALI-3+	1
SN65HVD76DGSR	RS485 Full Duplex Transceiver	IC7, IC8	SOP50P490X110-10N	SN65HVD76DGSR	2
MT29F4G01ABAFDW B-IT_F	Integrated Circuit	IC9	SON127P800X600X65- 8N	MT29F4G01ABAFDW B-IT_F	1
Pin header 1x3		J1	HEADER 3	Pin header 1x3	1
205338-0006	205338-0006 Connector Shrouded Header Positive Lock 6 Position 2mm Pitch Right Angle Surface Mount 1/8"	J2	FP-2053380006-MFG	CMP-00257- 106204701-1	1
G125-MH10605M4P	Connector	J3, J5, J6	G125- MH10605M4P_1	G125-MH10605M4P	3
901-144-88FX	Connector	J4	90114488FX	901-144-88FX	1
270nH +/-1% ?DCR	Inductor SMD 0603	L1	IND_0603	Inductor-SMD-RF- 0603	1
22nH	Inductor SMD 0402	L2, L3, L4	IND_0402	Inductor-SMD-RF- 0402	3
18nH	Inductor SMD 0402	L5	IND_0402	Inductor-SMD-RF- 0402	1
20nH	Inductor SMD 0603	L6	IND_0603	Inductor-SMD-RF- 0603	1
25nH	Inductor SMD 0603	L7	IND_0603	Inductor-SMD-RF- 0603	1
62nH	Inductor SMD 0402	L8	IND_0402	Inductor-SMD-RF- 0402	1
72nH	Inductor SMD 0402	L9	IND_0402	Inductor-SMD-RF- 0402	1
SMD 0805	LED YELLOW CLEAR 0805 SMD	LED1, LED2	FP-SM0805VCL-MFG	CMP-35283-000007-1	2
10k	Resistor SMD 0603 RATED	R1, R2, R3, R5, R6, R18, R20, R21	RES_0603	Resistor-SMD-0603	8
3.48k	Resistor SMD 0603 RATED	R4	RES_0603	Resistor-SMD-0603	1
4.7k	Resistor SMD 0603 RATED	R7	RES_0603	Resistor-SMD-0603	1
500	Resistor SMD 0603 RATED	R8, R12, R22, R23	RES_0603	Resistor-SMD-0603	4
5k	Resistor SMD 0603 RATED	R9, R10, R11, R13, R14, R15	RES_0603	Resistor-SMD-0603	6
120 1%	Resistor SMD 0603 RATED	R16, R17, R19	RES_0603	Resistor-SMD-0603	3
SI4464-B1B-FMR	High-Performance, Low-Current Transceiver, 119 to 960 MHz, -20 to 20 dBm Output Power, - 40 to 85 degC, 20-pin QFN, Tape and Reel	U1	SLAB-QFN-20- 2600x2600TP_M	CMP-0520-00036-1	1
	SENSOR DIGITAL -55C 125C 6WSON	U2, U3	FP-DRV0006A-IPC_C	CMP-04948-000045-1	2
TCBT-14+	Undefined or Miscellaneous	U4	TCBT14bias	TCBT-14+	1
TG2016SMN_28.0000 M-MCGNNM3	Crystal or Oscillator	Y1	TG2016SMN25000M MCGNNM3	TG2016SMN_28.0000 M-MCGNNM3	1
AB506-32.7680KHZ-1-T	CRYSTAL 32.7680KHZ 12.5PF SMD	Y2	FP-AB506-MFG	CMP-27762-012387-2	1
ABM3BAIG- 16.000MHZ-12-3-T	Crystal or Oscillator	Y3	ABM3BAIG	ABM3BAIG- 16.000MHZ-12-3-T	1